

BANA 3363

Spring 2025 (Wednesday Night)

Exam 1 Information and Review

Every topic you are responsible for knowing on the exam is in this document. You REALLY should look at it carefully.

Exam Information

Format

- A detailed list of topics is provided [below](#). If a topic is *not* on that list, it won't appear on the exam.
- The exam will consist of two parts:
 - a **multiple-choice** section with 20 - 25 questions completed on Blackboard using the classroom computers (**only one attempt is allowed**).
 - one or two (1 or 2) sort-answer/work-out problem(s) (which could contain several parts) that will be completed on a sheet of paper that I provide (**only one attempt is allowed**).
- All parts of the exam must be started and completed in class.
- The exam period will be **6 p.m. – 7:30 p.m. Central time** in our usual classroom. You will have the full 90 minutes to complete the exam. **There will NOT be a lecture afterward.**

Things You Will Need/Are Allowed to Use

- The formula sheets and statistical tables available in the course Blackboard page under the “Module 0: Syllabus, Textbook, & Course Information: (Start Here!)” content area. (**I will provide these, and you must use the copies I provide**).
- Excel on the **classroom computer only**. You cannot use a laptop, tablet, or other personal device. **Every question will be answerable without using Excel, however.**
- A **non-cellphone-based calculator** that cannot communicate wirelessly with any other device. You may use a graphing calculator (such as a TI-83, -86, or -89) or Excel, but that will be far more functionality than you will need. A calculator that can calculate exponents, logarithms, and square roots, such as [a TI-30](#), is all you need.
- **One (1) 8.5” x 11”** (standard sized) sheet of paper (“note sheet”) with anything you want hand-written (NOT typed) on both sides of it.

Based on years of teaching experience, I have the following advice on making this sheet:

- Don't try to set a world record for "Largest Amount of Mathy Things Crammed onto a Single Piece of Paper." **You won't have time to look up every single answer, and the questions on the exam aren't exact copies of problems worked in class or those on the review.** A note sheet is best used to **double-check** your answers.
- Make your own note sheet. **You are technically allowed to share note sheets** (for example, in study groups) **prior** to the exam (but **NEVER during the exam**), but **it is a very BAD idea to completely rely on someone else's work.** You will spend too much time trying to find things and you may not understand some of what is written because it wasn't written for your brain. Again, my advice is to try to understand why certain formulas are used in certain ways rather than trying to memorize what to do.

Things You **CANNOT** Use:

- Help from any person in any way, whether they are enrolled in the class or not.
- Notes (except the single-page note sheet as mentioned above). **PowerPoints, videos, discussion board posts, textbooks, or any other online or offline materials are NOT allowed.**
- Websites, cell phones, smart watches, tablets, the astral plane, Ouija boards, remote viewing, or any other online or offline source of information.
- Any form of artificial intelligence (or AI) system, including but not limited to ChatGPT.
- Any part of any solution that you may find in a textbook, from another student, or from one of the many cheating "homework help" websites.

A student found using any of the above prohibited items or engaging in prohibited activity will receive, at minimum, a zero (0) on the exam. Additional penalties may be imposed depending on the circumstances.

Exam Policies (Read **VERY** Carefully)

The exam is an **individual assignment**. **You MAY NOT work or consult with any other person(s) in any capacity, use any resources other than what is listed above, use any online or offline textbook, or in any way receive any kind of assistance with this exam.** Specifically:

- You **may NOT** work with any other person, student or not, for any part of the exam, even if you are just asking for clarification.
- You **may NOT** reproduce any part of the exam anywhere, either by hard copy, text message, email, screenshot, discussion board or internet posting, slam poetry session, interpretive dance, or any other means.

Doing any of the above will be considered cheating and will result, at a minimum, in a grade of zero (0) on the exam for all parties involved. Other penalties, in accordance with the Sam Houston State University Academic Dishonesty policy, may also result depending on the severity of the dishonesty.

- I will provide the formula sheet and statistics tables. **You cannot use your own copies of these items.**
- Both parts of the exam—multiple choice and short answer—**must be taken in class** between 6 p.m. and 7:30 p.m. on the day of the exam (**Wednesday**). **You CANNOT take the Blackboard portion of the exam anywhere but in person, and in our classroom.** I will take attendance to ensure everyone who submits the Blackboard portion was, in fact, in class, to do so.
- **You may not begin the exam late if any part of the exam has been submitted by any student.**

For example, if there is one submission of the multiple-choice portion of the exam on Blackboard by 6:30 p.m., you will **not** be permitted to take the exam.

There are no exceptions to any of the exam policies unless you have a documented accommodation from Services for Students with Disabilities(SSD):

<https://www.shsu.edu/dept/disability/>

Work-Out Problem(s) Information

- There will be one or two (1 or 2) problems, **which could contain several parts**, in which you will have to perform some calculations, write out the answers, and give interpretations. On these problems, I am far less concerned with seeing the “correct” answer than I am with seeing that you know how to get that answer. **In fact, if your setup is correct, but the calculation is wrong, you will receive most or all the points for that question!** On the other hand, **if you give the “correct” numerical answer but do not show work, you will get zero (0) points, even if that numerical answer is correct!** Although you can use a calculator on the exam, **writing “I plugged the following things into my calculator and here’s the output,” or something similar, DOES NOT count as work, and thus will be worth zero (0) points.**

There is one exception: **even if the calculated answer is wrong, it must still make sense in the context of the problem.** For example, if I ask you to calculate a 95% confidence interval for the mean waiting time in minutes in a drive-through lane, the lower and upper bounds should both be positive numbers since a negative wait time makes no sense.

- **One BIG Mistake to Avoid in Short Answer Problems**

When you write answers to short-answer problems, **be careful when you use the equal sign, =.** The things on either side of the = sign **must really be exactly equal to each other.**

Here’s an example of something that is **WRONG**:

We want to make a 95% confidence interval for a mean with a sample of 20.
Therefore,

$$\begin{aligned}1 - \alpha &= 0.95 \\&= 0.05/2 \\&= 0.025 \\&= t_{0.025,19} = 2.093.\end{aligned}$$

Why is this wrong? First, **0.95 is not equal to 0.05/2** (although it is true to say that $0.05/2 = 0.025$). **Just ask yourself, is $0.95 = 0.025$? No! They aren't the same thing; thus, they aren't equal, thus we can't use the equal sign.**

Second, **0.025 is not equal to 2.093** (although it is true that $2.093 = t_{0.025,19}$.)

It's always a good idea to read back over your work to make sure that what you've written makes sense in the real world, because the goal of statistics is to explain the real world.

- Partial credit is possible for most or all parts of the short-answer question, so you should always write *something* down, even if you aren't sure. **You cannot get negative points on a short-answer question, so always try.** To not even attempt the short-answer question would be like a player *holding onto* the basketball at the buzzer instead of taking the shot!
- If you make a mistake on a multi-part problem for which later parts rely on statements or calculations from earlier parts, as long as you are consistent in that mistake throughout the problem, you will not be penalized multiple times.

For example, suppose you are given a hypothesis testing problem with a sample size $n = 30$ that should be tested at $\alpha = 0.05$ and the hypotheses should be

$$H_0: \mu = 9$$

$$H_1: \mu \neq 9$$

(that is, a two-sided test), but you incorrectly specify the hypotheses in Step 1 as

$$H_0: \mu \leq 9$$

$$H_1: \mu > 9$$

(that is, a right-tailed test). Then, to be consistent with Step 1, you specify the rejection region in Step 3 to be a right-tailed rejection region: all $t > t_{0.05,29} = 1.699$ (when it should actually be all $t < -t_{0.025,29} = -2.045$ and all $t > t_{0.025,29} = 2.045$ for a two-sided test). Then, you would lose points **only once** on this problem for the incorrect statement of the hypotheses. You would

not be penalized for getting the rejection region incorrect since it would have been correct if your hypotheses statement were correct.

Study Guide

You are responsible for understanding all of the topics below; that is, you should prepare to be able to work on any of the topics below. If you don't understand something in the review, it is YOUR responsibility to ask for help.

However, because the exam has a time limit, not all of these topics will appear on the exam. Here's the rule for any of my exams:

If a topic DOES NOT appear in the list of topics below, it WILL NOT appear on the exam. If a topic appears below, it COULD appear on the exam.

I write all of the questions on my exams, Blackboard assignments, and examples in the Course Packet myself; I hardly ever use questions from test banks, and even then, I change most of the question text anyway. Therefore, the best way to get an idea of how I could ask a question is to review the examples from the relevant videos in Blackboard and the Blackboard assignments. **Trying to find a test bank online to help you study will be a BAD IDEA.** First, it's cheating, which is simply wrong (yes, even in a class that isn't in your major), and such behavior is prohibited by Sam Houston State explicitly in its policies on academic dishonesty (see the syllabus). Second, it will be a waste of time that you could spend, you know, actually *studying*!

Module 1: Introduction to Statistics and Review of BANA 2372

- Do you know the definitions of the terms “population” and “sample”?

A population refers to the whole collection of persons, places, objects, or events (i.e., “things”) that we wish to learn about.

The term “population” can be misleading: it suggests that such a collection is fixed over time and space—often not true! Usually, we are really interested in an underlying process that produces a specific collection of objects at any given time.

Example:

We want to estimate the true mean hourly wage (including tips) of servers at steakhouses in Texas. We could say the “population” is all servers in Texas who work at steakhouses. But this population changes moment by moment: people get hired, quit, get promoted, or get fired over the course of every day. What we usually care about is not the fixed collection of servers *right at this moment*, but the process of working such a job in Texas.

Whether we think about the object of study being a population or a process, a **sample** is a collection of items gathered from the population or process that we use to make inferences, or “guesses” about population **parameters**, such as the mean or proportion.

- Do you know what “parameters” and “statistics” are?

A **parameter** is a numerical characteristic of a population or process. These are the unknown values that we are trying to estimate using a sample. A **statistic** is a descriptive measure of a sample which may be used to “guess at” the value of the parameter.

Here’s an example in Figure 1 that shows the relationship. Suppose we wanted to determine what the true mean miles-per-gallon (MPG) a Toyota Prius driver could expect in real-world driving conditions. The **population** is all Priuses ever made (or that will ever be made; this is an example of an “infinite population”). The **parameter** is μ , the population mean MPG. To do so, we could take a **sample** of $n = 4$ Priuses and calculate the **sample mean**, $\bar{y}$, and sample standard deviation, s .

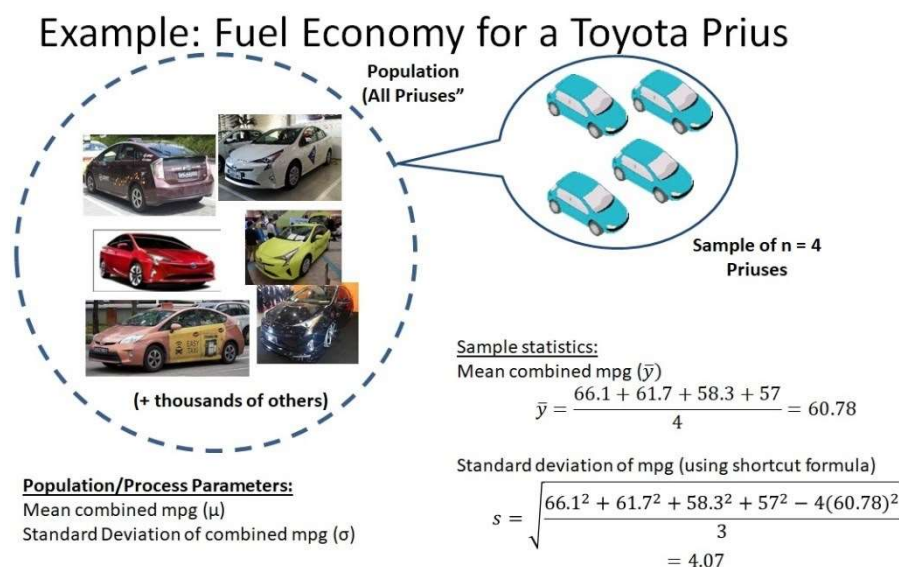


Figure 1: Example of sampling from a population

- Do you know the difference between descriptive and inferential statistics?

Descriptive statistics is a collection of techniques for organizing, summarizing, and presenting a **sample** of data in a convenient, informative way. **Inferential statistics** is a collection of techniques designed to use sample data to make generally applicable statements about the natural process or **population** from which the data arose.

- Do you know how to calculate a sample mean and sample standard deviation?

For example, a sample of $n = 4$ Prius vehicle miles per gallon (MPG) ratings were the following:

66.1, 61.7, 58.3, 57

The sample mean MPG, symbolized by $\bar{y}$ (pronounced “y bar”) is the sum of these numbers divided by the number of observations. The sample standard deviation can be calculated using the “short-cut formula” shown below:

$$s = \sqrt{\frac{\sum y_i^2 - n\bar{y}^2}{n - 1}}$$

The textbook, definitional formula for the sample standard deviation is:

$$s = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n - 1}}$$

and is shown in Figure 1 on page 6.

- Do you know all the characteristics of **any** normal distribution?
 - A normal distribution is defined by its mean, μ , and standard deviation, σ . Given those two values, if we wanted to, we could plot the normal curve using, say, a graphing calculator, and we would get a bell-shaped curve (you won’t have to plot the curve on an exam, however).
 - We can write the phrase “a normal distribution with mean μ and standard deviation σ ” in abbreviated form as $N(\mu, \sigma)$.
 - Review the other properties of the normal distribution under “The Normal Distribution” material Module 1 on Blackboard.
- Do you know what the standard normal distribution is?

The standard normal (Z) distribution is one particular normal distribution; there are (infinitely) many other normal distributions, each one specified, as noted above, by its mean, μ , and standard deviation, σ . These are parameters and are like the “DNA of a normal distribution.” Think of it this way: every human being has certain traits in common. The normal distribution is like a human; many humans exist. You, dear student, are a specific human who has those common human traits, but who also has traits specific to you which are a part of your DNA. Humankind in general is like a normal distribution; you, as a specific individual, are like the standard normal distribution.

The standard normal distribution is just one particular normal distribution with a specific mean (do you know what it is?) and standard deviation (do you know what it is?)

- Do you know “the **Z-score theorem**” that lets us “transform” any normal random variable with mean μ and standard deviation σ into a standard normal (Z) random variable?

I may have phrased this as the rule that lets us “turn a question about Y into a question about Z.” We used it throughout the material on the normal distribution.

The rule is: if Y has a normal distribution with mean μ and standard deviation, σ , then

$$Z = \frac{Y - \mu}{\sigma}$$

has a standard normal distribution, which is a normal distribution with mean = 0 and standard deviation = 1. (Oh, it looks like I gave you the answer to the previous bullet point!)

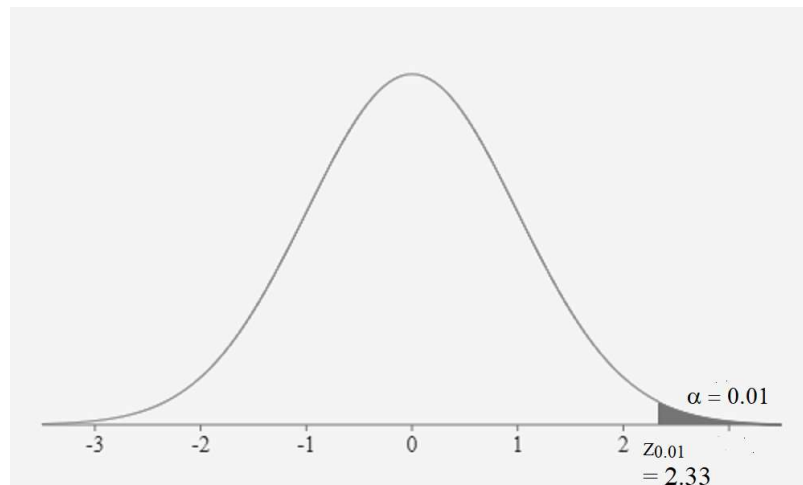
- Suppose a variable (e.g., weight, height, IQ, blood pressure, dollars, whatever...), Y, has a normal distribution with mean μ and standard deviation σ . Do you know how to work with the normal distribution in the following ways?
 - Find probabilities of the “less than,” “greater than,” and “between” types:
 - $P(Y < y)$ [this is numerically equivalent to $P(Y \leq y)$]
 - $P(Y > y)$ [this is numerically equivalent to $P(Y \geq y)$]
 - $P(y_1 < Y < y_2)$, where y_1 and y_2 are two specific numbers.
[this is numerically equivalent to $P(y_1 \leq Y \leq y_2)$]
 - Find a value on the standard normal (i.e., Z) distribution with a specified probability (or area; **area IS probability**) of α to the right. Recall that we indicated this value using the notation z_α . **This notation ALWAYS refers to this interpretation.**

Example:

Find $z_{0.01}$.

Solution:

By definition, " $z_{0.01}$ " is "the value of the standard normal that has area of 0.01 to the right." Drawing a picture can help:



If there is area of 0.01 to the *right*, there must be an area $1 - 0.01 = 0.99$ probability to the left. Since the z table always gives us probability to the left, we look up 0.99 inside the table and read out to the row and column. The closest we can get is 0.9901. Reading out to the row and column, we see that the z score that gives this probability is 2.33, so $z_{0.01} = 2.33$.

- Find an interval on the standard normal (i.e., Z) curve that contains a specific probability (area) of $1 - \alpha$? That is, find $z_{\alpha/2}$ such that

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha.$$

For example, if we wanted to find an interval on the standard normal distribution that contains probability (which is, again, the same thing as area) of 0.95, as shown below, we would proceed as follows:

$$1 - \alpha = 0.95$$

$$\alpha = 0.05$$

$$\text{Use } \pm z_{\alpha/2} = \pm z_{0.05/2} = \pm z_{0.025}$$

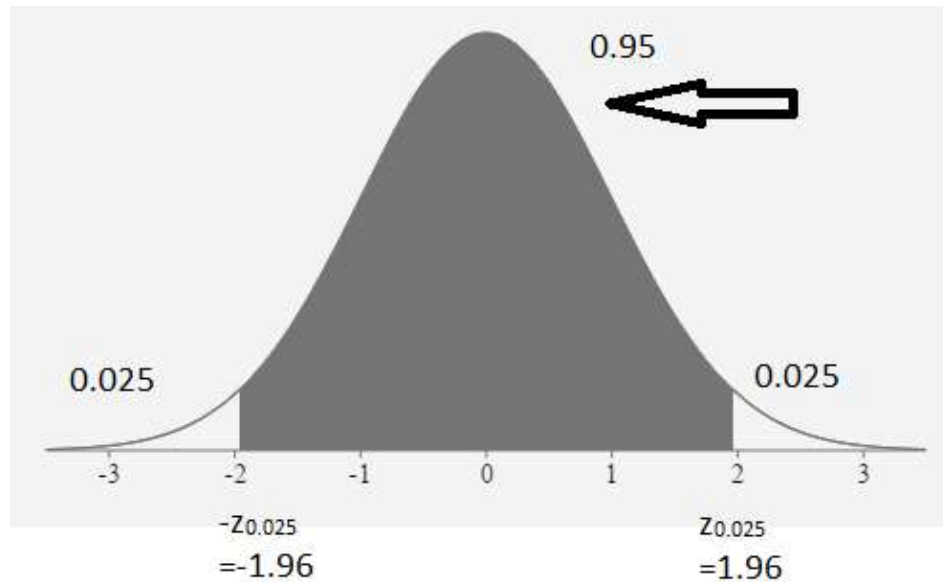


Figure 2: The standard normal distribution with the 95% region (-1.96 to 1.96) shaded.

Thus, the interval that gives us 95% area in the shaded region is from -1.96 to 1.96 . That is,

$$P(-1.96 \leq Z \leq 1.96) = 0.95.$$

Because of the interpretation of z-scores (see above), this says that 95% of any normal distribution lies within 1.96 standard deviations around the mean. This is where the “95” part of the 68-95-99.7 rule comes from, which states that about 95% of any normal curve lies within 2 standard deviations of the mean. The “2” is just 1.96 rounded up.

Module 2: The Basics of Inference: Confidence Intervals and Hypothesis Tests

Note: In general, the sample mean is denoted by a letter with a bar over it, as in $\bar{X}$ or $\bar{Y}$. I’ve used both. The textbook uses $\bar{X}$ mostly. The specific letter isn’t important; anything with a “bar” over it refers to the sample mean.

- Do you know what a sampling distribution is and how it arises?

A nice demonstration can be found here: http://onlinestatbook.com/stat_sim/sampling_dist/

That’s a link to the program I showed (either in class or in a video, depending on whether you’re taking this class online or face-to-face) with the blue and black boxes that dropped down and formed a pattern.

- Do you know what the mean and standard deviation are of the sampling distribution of the sample mean, $\bar{Y}$?

See Section 7.5 of the MindTap textbook, equations 7.1 and 7.2.

- Do you know what the Central Limit Theorem states?

This bullet and the one above it are really important. Together, they tell you about what the (approximate) distribution of the sample mean from any population will look like. See the sampling distribution link above, and Section 7.5 of the MindTap textbook.

- Do you know how to find probabilities for the sampling distribution of the sample mean using the central limit theorem? That is, do you know how to turn questions about $\bar{Y}$ into questions about Z ?

This involves using the z-score theorem again. The big change is remembering to divide by the correct standard deviation. Remember that the standard deviation of the sampling distribution of the sample mean is $\sigma_{\bar{Y}} = \frac{\sigma}{\sqrt{n}}$.

Module 2 Part 1: Confidence Interval Basics/Confidence Interval for One Mean

- Can you construct, and interpret in context, a confidence interval for **one mean**?

On the exam, **all of the intervals for means should be constructed using values from Student's t distribution** because we don't want to assume that we know σ , the population standard deviation. We only ever really know s , the sample standard deviation.

*Note: σ ("sigma") is **NOT** the same thing as s . In fact, s is an estimate of σ , just like $\bar{Y}$ is an estimate of μ .*

Example:

Suppose I am working in quality control for a company making 32-pound buckets of powder and calculate a sample mean weight on a sample of six buckets to be $\bar{y} = 31.8$ pounds, with a (sample) standard deviation $s = 0.23$ pounds. Then, I can "focus" on this data and say what values are "good guesses" (i.e., that give me, say, 95% confidence) for the true but unknown mean weight, μ . I get those good guesses using the confidence interval formula:

$$\bar{y} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

For 95% confidence and a sample of $n = 6$ buckets, I look in the Student's t distribution table at row $6 - 1 = 5$ and column 0.025, and I get $t = 2.571$. Plugging things in, the interval is

$$31.8 \pm 2.571 \frac{0.23}{\sqrt{6}},$$

which gives an interval of 31.56 to 32.04 (the lower bound is 31.56 and the upper bound is 32.04).

I would then say that I am 95% confident (where “confidence” is based on the method used to calculate the interval) that the true mean weight of the powder is between 31.56 and 32.04 pounds.

The range of numbers between 31.56 and 32.04 is my range of “good guesses” for the true mean weight of the buckets that the process is making.

- Using a confidence interval, could you determine whether a particular value of the parameter was a “reasonable guess”?

For example, suppose I asked you if it was reasonable that the true mean fill of the process in the above example was 32 buckets using the interval above. Is it?

Yes! Why? Because 32 lies in the interval of “good guesses,” that is, the confidence interval. So, it is a good guess.

What about if I asked you if it was reasonable to conclude that the true mean weight was 35 pounds?¹

Module 2 Part 2: Hypothesis Testing Basics/Hypothesis Test for One Mean

- Do you know the general ideas behind hypothesis tests? For example, do you know what “null” and “alternative” hypotheses are? Do you know what a test statistic is? Do you know what a Type I (“false alarm”) error is? Do you know what conclusions you can make about a parameter based on performing a hypothesis test?

Example: A dean of a college of business is concerned about a recent claim in *The Boston Globe* that today’s college students study less (in terms of hours per week) than students 50 years ago. Specifically, the article claims that today’s students spent a mean time of 14 hours per week studying. The dean decides to determine if there is evidence that the mean study time, μ , of students in his college is different from the number claimed in the article. After gathering a random sample of 35 students of various majors in the college of business, he found that the sample mean time spent studying per week was $\bar{y} = 16.4$ hours, with a standard deviation $s = 7.2$ hours.

Answer the following questions about this situation.

¹ This would **not** be a reasonable conclusion because the **confidence interval does not** include “35.” Basically, the confidence interval has told us the range of “good guesses” based on the data, and that range doesn’t include the “guess” of 35, so it’s not a good guess, based on the data.

Q). If the dean decides to use a hypothesis test to address this question, what general steps will he take in order to follow the hypothesis testing procedure?

A). The four steps common to all hypothesis tests are: define the hypotheses, calculate the test statistic, determine the rejection region, and make a conclusion.

Q). What are the two types of hypotheses used in hypothesis testing?

A). The two hypotheses are the null hypothesis, H_0 , pronounced “h not” or “h sub-zero,” and the alternative hypothesis, H_1 (the MindTap textbook writes the alternative hypothesis as H_a , but both refer to the same thing).

Q). What would be the null and the alternative hypotheses for examining the mean study time in the example above?

A). The two hypotheses must be mutually exclusive, which means, roughly, that they are “opposites” of each other. If H_1 is true, then H_0 must be false; if H_1 is false, H_0 must be true.

Most often, the null hypothesis H_0 is a claim that the current state of nature is equal to something, or that the current state of nature hasn’t changed. In the context of examining study time, the article claims that the true mean amount of time that students spend studying per week is equal to 14. This is the null hypothesis.

Using symbols, we would write

$$H_0: \mu = 14$$

The alternative hypothesis H_1 is then the “opposite” of the statement above. Well, the opposite of the situation where the mean equals 14 is that the mean **does not equal** 14. Using symbols, we can write

$$H_1: \mu \neq 14$$

(note that the symbol $\neq$ is read as “not equal to”).

We typically write these together to perform Step 1 of the hypothesis testing procedure, “define the hypotheses.” So, we would write

$$\begin{aligned} H_0: \mu &= 14 \\ H_1: \mu &\neq 14 \end{aligned}$$

Q). How would conducting a hypothesis test be different than calculating a confidence interval for the true mean study time?

A). A confidence interval would look at the information given in the data (i.e., the sample mean of 16.4 and sample standard deviation of 7.2 hours) and determine a range (i.e., an **interval**) of “good guesses” or “reasonable guesses” about the true mean study time μ might be.

A hypothesis test looks at the situation from another angle. The test will allow the dean to see if a **particular guess** at the true mean is “good” or not. The particular guess he will look at is 14, the claim made by the article in *The Boston Globe*.

Q). In the context of the problem above, what would a Type I error be?

A). A Type I error can be thought of as a “false alarm,” that is, **rejecting the null hypothesis when the null hypothesis is actually true**. For example, a Type I error would be made if someone took a pregnancy test and it came up positive, +, **when they really were not pregnant** (that is, the true state of nature is negative).

This table can make things easier to visualize:

		Decision	
Reality		Reject H_0	Fail to Reject H_0
	H_0 true	Error (Type I)	Correct Decision 😊
	H_0 false	Correct Decision 😊	Error (Type II)

Figure 3: Decisions in hypothesis testing. A Type I error occurs when we reject H_0 when H_0 is true.

In the context of study time, a Type I error would occur if the dean **concluded that the true mean study time was not 14 hours per week when it really was 14 hours per week**.

The consequences of a Type I error depend on what application we are dealing with. Making a Type I error in the context of study time could result in the dean encouraging faculty to require more homework and projects in more classes so that this average increases. This is only one of the possible actions the dean could take.

Q). What are the only two conclusions that the dean could make after conducting a hypothesis test?

A). In general, the **only** two conclusions you can make in a hypothesis test are either “**reject the null hypothesis**” (“**reject H_0** ”) or “**fail to reject the null hypothesis**” (“**fail to reject H_0** ”).

If the phrase “fail to reject” is hard to wrap your head around, you can also say “cannot reject the null hypothesis,” “unable to reject the null hypothesis,” “not enough evidence to reject the null hypothesis,” etc. The key, though, is that our decision is about the **null** hypothesis. Either there is evidence in the data that the null hypothesis is false, or there is not.

You might see the phrase “accept the null hypothesis” in some online resources or in textbooks, but this phrasing is very misleading. **Not rejecting the null hypothesis is different from concluding that the null is true.** Our conclusion in this case is kind of wishy-washy; all we can say is that we don’t have enough evidence to conclude that the null is not true. In other words, there is not sufficient evidence to conclude that the null is false.

- Can you define the basic terms of hypothesis testing and construct a hypothesis test for **one** population **mean**?

Note that when I say, “construct a hypothesis test,” I will expect you to be able to go through all of the four steps:

- 1). Define (or Specify) the hypotheses
- 2). Calculate the test statistic
- 3). Determine the rejection region²
- 4). Make a conclusion in the context of the problem (reject or fail to reject the null hypothesis, and state what that decision means in the context of the problem).

For testing means (and proportions), there are three types of hypotheses:

- right-tailed
- left-tailed
- two-tailed (or “two-sided”)

Right-tailed and left-tailed tests are called “one-tailed” or “one-sided” tests. The type of alternative hypothesis specified in Step 1 determines how the rejection region (or p-value) in Step 3 is determined. This basic principle applies to one-mean, two-mean, one-proportion, and two-proportion tests.

Example: Right-Tailed Test

An advertising agency is working on some commercials that it plans to air for a client during the next Super Bowl. Because the ads cost so much (millions of dollars for just a

²Recall that we can also calculate a “p-value” to help us decide a hypothesis test. Because p-values require software to calculate in general, I will **not** ask you to **calculate** a p-value, but I could ask you about what a p-value tells you.

30-second spot), it is important for scheduling purposes to determine how many commercials, on average, a viewer will watch, from beginning to end, over the course of the game. The advertiser specifically wants to determine if the true mean number of Super Bowl commercials watched is more than 15. Using a random sample of 114 viewers from last year's game, she found that the sample mean number of commercials watched was 15.3, with a standard deviation of 5.7. At the 0.05 level of significance, test the hypothesis that the advertiser would be interested in. Show all four steps of the hypothesis testing procedure.

Solution:

1). Define (or Specify) the hypotheses

Because the advertiser is interested in seeing if the true mean is "more than 15," the test is a one-tailed (specifically, right-tailed test), and the hypotheses are as follows:

$$\begin{aligned}H_0: \mu &\leq 15 \\H_1: \mu &> 15\end{aligned}$$

2). Calculate the test statistic

The test statistic is a Student's t (**we always use Student's t for problems involving means**). It is given by the following formula:

$$t = \frac{\bar{y} - \mu_0}{s / \sqrt{n}}$$

Using the information from the problem, the test statistic is

$$\begin{aligned}t &= \frac{15.3 - 15}{5.7 / \sqrt{114}} \\&= 0.562\end{aligned}$$

3). Determine the rejection region (or compute the p-value)

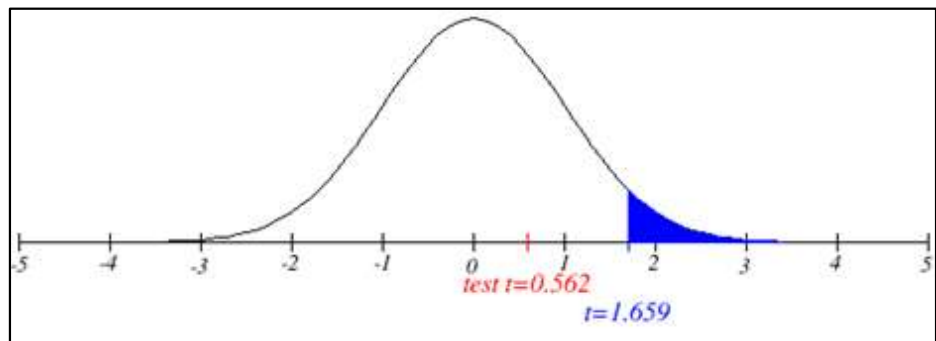
Rejection Region Approach

Because the alternative hypothesis, H_1 , has a ">" sign, the rejection region will be in the right tail of the Student's t distribution. So, we need to find $t_{\alpha, n-1}$. Because $\alpha = 0.05$ from the problem statement, the rejection region is all $t > t_{0.05, 113}$ (the 113 comes from the fact that the degrees of freedom are $n - 1$, so $114 - 1 = 113$). In words, $t_{0.05, 113}$ is "the value of the Student's t distribution with 113 degrees of freedom and 0.05 probability to the right."

Notice in the Student's t table, we don't have a 113 row. We have a row for 110 and a row for 120. Since 113 is closer to 110 than 120, we will use row 110 (you can see there's hardly any difference between them). Notice, too, that we need the second column from the right, since that gives us $t_{0.05}$. Therefore, we will use $t_{0.05, 113} = 1.659$ to mark the rejection region, and therefore the rejection region is all $t > 1.659$.

110	1.289	1.659	1.982	2.361	2.621
120	1.289	1.658	1.980	2.358	2.617
130	1.288	1.657	1.978	2.355	2.614
140	1.288	1.656	1.977	2.353	2.611

Here's a picture; the rejection region is shaded.



(If you are in an online class, you wouldn't have to draw the rejection region; you would just specify it, as I have done).

P-Value Approach

Without software, you cannot compute the p-value. **But you could be asked about what the p-value would have to be based on your decision to reject the null hypothesis.** This is because the p-value approach and the rejection region approach will always lead to the same conclusion for a given value of α .

Remember, the p-value rule:

*If the $p\text{-value} \leq \alpha$, **reject** the null hypothesis. Otherwise, fail to reject the null hypothesis.*

4). Because $t = 0.562 < 1.659$ (that is, since the test statistic falls **outside** of the rejection region), we **fail to reject** the null hypothesis. **In context**, this says that we **cannot conclude** that the true mean number of commercials that viewers watch all the way through during the Super Bowl is larger than 15. We can't say what it is, just that it's some number less than or equal to 15.

Because we **failed to reject the null hypothesis**, we would **know** that the **p-value** (whatever its actual value) **must be greater than 0.05**, *because* of the p-value rule above.

Example: Left-Tailed Test

A real estate developer in Houston wants to see if the mean price for a single-family home in a particular neighborhood has decreased from what it was before the Great Recession. The mean price before the Great Recession was \$191,300. In a sample taken this year of 123 homes, it was found that the mean was \$178,614 with a standard deviation of \$80,702. At the $\alpha = 0.10$ level of significance, test the hypothesis the real estate developer would be interested in. Show all four steps of the hypothesis testing procedure.

Solution:

1). Define (or Specify) the hypotheses

Because the advertiser is interested in seeing if the mean price has *decreased* from what it was before the Great Recession, the hypotheses will be as follows:

$$H_0: \mu \geq 191,300$$

$$H_1: \mu < 191,300$$

2). Calculate the test statistic

The test statistic is a Student's t. **The formula for the test statistic is always the same, regardless of whether the test is one or two-tailed.** It is, specifically,

$$t = \frac{\bar{y} - \mu_0}{s / \sqrt{n}}$$

Using the information from the problem, the test statistic is

$$t = \frac{178,614 - 191,300}{80,702 / \sqrt{123}}$$

$$= -1.743$$

3). Determine the rejection region (or compute the p-value)

Rejection Region Approach

Because the alternative hypothesis, H_1 , is a “less than” type, the rejection region will be in the left tail of the Student’s t distribution. So, we need to find $-t_{\alpha, n-1}$, which is just the value in the Student’s t table with area of α to the right with $n - 1$ degrees of freedom multiplied by -1 to put the rejection region in the left tail. **This works because of the symmetry of the Student’s t distribution around 0.** Because $\alpha = 0.10$ from the problem statement, the rejection region is all $t < -t_{0.10, 122}$ (the 122 comes from the fact that the degrees of freedom are $n - 1$, and $123 - 1 = 122$). In words, $-t_{0.10, 122}$ is “the value of the Student’s t distribution with 122 degrees of freedom and 0.10 probability to the left.”

Notice in the Student’s t table, we don’t have a 122 row. We have a row for 120 and a row for 130. Since 122 is closer to 120 than 130, we will use row 120. Notice, too, that we need the first column from the right because that gives us $t_{0.10}$. Therefore, we will use $-t_{0.10, 122} = -1.289$ to mark the rejection region, and therefore the rejection region is all $t < -1.289$.

110	1.289
120	1.289
130	1.288
140	1.288

P-Value Approach

Without software, you cannot compute the p-value. **But you could be asked about what the p-value would have to be based on your decision to reject the null hypothesis.** This is because the p-value approach and the rejection region approach will always lead to the same conclusion for a given value of α .

4). Because $t = -1.743$, and $-1.743 < -1.289$, (that is, since the test statistic falls inside of the rejection region), we reject the null hypothesis. **In context**, this says that we can conclude that the true mean selling price of houses in this particular Houston

neighborhood is less than what it was before the Great Recession. **(Of course, this is assuming that the sample of 123 house prices could be considered a reasonable representation from the “population” (or the “house-price setting process”) behind all houses in the neighborhood.**

Because we **rejected the null hypothesis**, we would *know* that the **p-value** (whatever its actual value) **must be less than 0.10**, which is the level of significance.³

Example: Two-Tailed Test

The age demographics of online shoppers are of interest to a local retailer that is considering putting more resources toward its online business. The company wants to know if the mean age of its customers who are online shoppers differs significantly from 23.3 years, which is the claim that a trade publication has made about shoppers in this demographic nationwide. After taking a sample of 36 current customers who self-identified as regular online shoppers, the company found the mean age was 24.2, with a standard deviation of 5.3 (both in years). Suppose the company wishes to test the hypothesis implied in the paragraph above at the $\alpha = 0.05$ level of significance.

Show all four steps of the hypothesis testing procedure.

Solution:

1). Define (or Specify) the hypotheses

Because the retailer is interested in seeing if the mean age has *changed* from what it was, the hypotheses will be as follows:

$$H_0: \mu = 23.3$$

$$H_1: \mu \neq 23.3$$

2). Calculate the test statistic

The test statistic is a Student's t. **The formula for the test statistic is always the same, regardless of whether the test is one or two-tailed.** It is, specifically,

³ Just for your information, if you were to use Excel (which is **NOT required but can be used**) you could find the p-value using the T.DIST function by typing the following into a blank cell:

=T.DIST(-1.743,122,TRUE)

This gives us a p-value of 0.0419, which confirms what the theory would tell us: the p-value is less than the level of significance (0.05).

$$t = \frac{\bar{y} - \mu_0}{s / \sqrt{n}}$$

Using the information from the problem, the test statistic is

$$t = \frac{24.2 - 23.3}{5.3 / \sqrt{36}}$$

$$= 1.019$$

3). Determine the rejection region (or compute the p-value)

Rejection Region Approach

Because the alternative hypothesis, H_1 , is of the “not equal to” type, the rejection region will be split in half between the upper and lower tails of the Student’s t distribution. So, we need to find $t_{\alpha/2, n-1}$. Because $\alpha = 0.05$ from the problem statement, the rejection region is all $t < -t_{0.025, 35}$ **and** all $t > t_{0.025, 35}$ (the 35 comes from the fact that the degrees of freedom are $n - 1$, and $36 - 1 = 35$). The level of significance, 0.05, is split in half, putting an area of 0.025 in each tail.

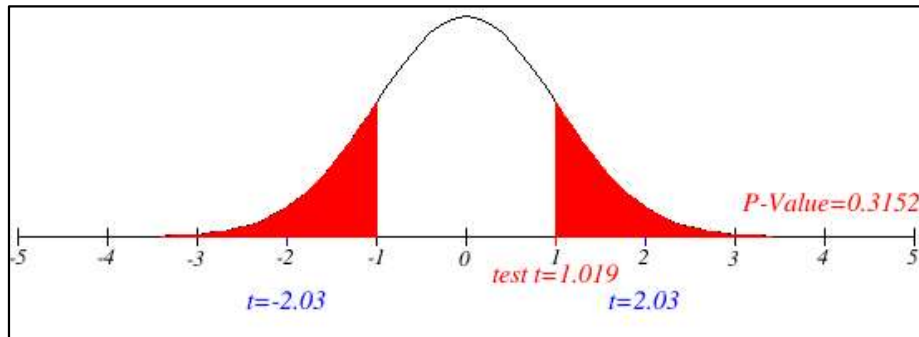
Looking at the $t_{0.025}$ column of the t table and the 35th row, we see $t_{0.025, 35} = 2.030$. Therefore, the rejection region is all $t < -2.030$ and all $t > 2.030$.

29	1.311	1.699	2.045
30	1.310	1.697	2.042
35	1.306	1.690	2.030
40	1.303	1.684	2.021
45	1.301	1.679	2.014

P-Value Approach

Without software, you cannot computer the p-value. **But you could be asked about what the p-value would have to be based on your decision to reject the null hypothesis.** This is because the p-value approach and the rejection region approach will always lead to the same conclusion for a given value of α .

We can see this relationship in the graph below



4). Because $t = 1.019$, which is not in either tail (that is, since the test statistic falls **outside** of the rejection region), we **fail to reject** the null hypothesis. **In context**, this says that the company **cannot conclude** that the mean age of its customers who identify as online shoppers is different from what was reported by the trade publication. (Of course, the BIG assumption here is that these 36 customers are truly representative of the online shopper population that the trade publication is reporting on. Before making any expensive changes, the company should make sure its sampling approach is reasonable).

Because we **failed to reject the null hypothesis**, we would *know* that the **p-value** (whatever its actual value) **must be greater than 0.05**, which is the level of significance.⁴

Module 2 Part 3: Confidence Intervals and Hypothesis Tests for Population Proportions

- Can you construct a confidence interval for **one** population proportion **and** interpret it in the context of the problem?

When working with proportions (making confidence intervals or constructing hypothesis tests), we **always** use the **standard normal** (that is, the **Z**) distribution. We use the Student's *t* distribution for **means** (see previous sections).

Example:

In a nationwide survey of 1000 college students, 650 of them responded that they would be interested in doing a study abroad program at some point in their college careers. Find a 95%

⁴ Just for your information, if you had access to Excel, you could find the p-value using the T.DIST.2T function by typing the following into a blank cell:

$$=T.DIST.2T(1.019,35)$$

This gives us a p-value of 0.3152, which confirms what the theory would tell us: the p-value is greater than the level of significance (0.05).

confidence interval for the true proportion of college students who feel this way **and interpret the interval in context.**

Answer:

The sample proportion is $\hat{p} = \frac{650}{1000} = 0.65$. The sample size is $n = 1000$.

We always use a value of Z (the standard normal distribution) for inferences on proportions, so we need $z_{0.025}$ for 95% confidence. This is the value of the standard normal distribution with 0.025 area to its right, or $1 - 0.025 = 0.975$ area to its left. Since the Z table always gives us probabilities less than, look up 0.975 inside the table and read out to the row and column to get $z_{0.025} = 1.96$.

(If you're not sure why, draw the standard normal curve and say to yourself, "If we want 95% of the normal curve to be between two points, there must be 5% not between those two points. Half of 5% is 2.5% or 0.025." Therefore, we need to find $z_{0.025} = 1.96$. See also Figure 2.

Then the 95% confidence interval is $0.65 \pm 1.96 \sqrt{\frac{0.65(1-0.65)}{1000}}$

Lower: 0.62

Upper: 0.68

Interpret in context: We are 95% confident, based on the method used to calculate the interval, that **the true proportion** of college students nationwide who are interested in study abroad is between 0.62 and 0.68 (or, you could say that between 62% and 68% of college students nationwide would be interested in a study abroad opportunity, with 95% confidence).

What if we changed the confidence level to 90%, or 99% or any other level? **All that changes is the value of Z that we use!**

For example, the 99% confidence interval would use $z_{0.005} = 2.575$ **(again, draw the picture as I did in class to see why we use this value of z for 99%)**. Then the interval would be

$$0.65 \pm 2.575 \sqrt{\frac{0.65(1 - 0.65)}{1000}}$$

Lower: 0.611

Upper: 0.689

Notice the interval is slightly wider because we want to be “more sure” of capturing the true, unknown parameter p .

- Do you know what a **margin of error** is?

In general, it’s just the stuff after the $\pm$ in a confidence interval. For the example above regarding study abroad, the margin of error for the 95% confidence interval was

$$\pm 1.96 \sqrt{\frac{0.65(1-0.65)}{1000}} = \pm 0.03.$$

For a 99% confidence interval, the margin of error would be larger:

$$\begin{aligned} \pm 2.575 \sqrt{\frac{0.65(1 - 0.65)}{1000}} \\ = \pm 0.0388. \end{aligned}$$

- Can you perform a hypothesis test for **one** proportion?

Note that when I say, “construct a hypothesis test,” I will expect you to go through all of the four steps (which are the same general steps as for working with a hypothesis test for a mean) are:

- 1). Define (or Specify) the hypotheses.
- 2). Calculate the test statistic.
- 3). Determine the rejection region (or compute the p-value).
- 4). Make a conclusion in the context of the problem (reject or fail to reject the null hypothesis, **and state what that decision means in the context of the problem**).

Example:

A dean of a college of business wants to see if the college should invest more in a study abroad program. The university’s Board of Regents will consider new funding only if there is evidence that over 40% of students in the college indicate that they would be interested in such a program. He takes a random sample of 300 business students and finds that 125 of them would be interested in a study abroad program. Test the hypothesis that the Board of Regents should consider new funding at the $\alpha = 0.10$ level of significance.

Answer:

We have the sample proportion $\hat{p} = \frac{125}{300} = 0.417$. The sample size is $n = 300$. The hypothesis test procedure would be as follows:

1). Define (or Specify or Determine) the hypotheses

Since the Board will consider new funding only if the true proportion of students interested is above 40% (or 0.40), this is a right-tailed test, and the hypotheses are:

$$H_0: p \leq 0.40$$

$$H_1: p > 0.40$$

2). Because we are dealing with a proportion, the test statistic is a z:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

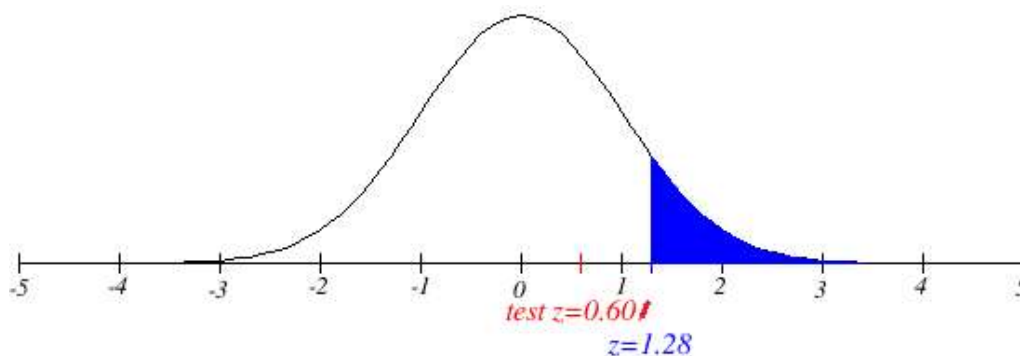
$$z = \frac{0.417 - 0.40}{\sqrt{\frac{0.40(1 - 0.40)}{300}}} = \frac{0.017}{0.0283} = 0.601$$

Notice that, when doing the test for proportions, we plug in the null hypothesis value of p , p_0 , in the denominator, not the sample proportion of 0.417, as we would do if this were a confidence interval.

3). Determine the rejection region (or compute the p-value)

Since the alternative hypothesis is pointing to the right (“greater than”), the rejection region is on the right. It’s in the standard normal distribution, marked by the value $z_{0.10}$ which, like the interpretation of the Student’s t value above, is short for “the value of the standard normal distribution with probability 0.10 to its right.” We need this value because the problem said the test should be done at the $\alpha = 0.10$ level. The value is found, as I’ve shown in class, by looking up $1 - 0.10 = 0.90$ in the Z table and reading out. You will find that the value $z_{0.10} = 1.28$.

Here is a picture:



As with the test of a mean—and with all hypothesis tests for that matter—you could not compute the p-value without software.

4). Make a conclusion in the context of the problem.

Because the test statistic value does not fall in the rejection region, **we fail to reject the null hypothesis. In context, we cannot** conclude that the true proportion of college students at the university who would be interested in study abroad is more than 40%. **Therefore, the Board of Regents will not consider new funding.**

Based on this decision, we know that the p-value—whatever it actually is—must be greater than 0.10, because of the decision to fail to reject.

Key General Concepts to Know for All Exams

- Do you know the **differences** and **similarities** between hypothesis tests and confidence intervals and when it would be appropriate to use each one?

Whether we are testing a mean or a proportion (or other types of hypotheses that we will discuss later), a **hypothesis test** is a procedure through which a **particular claim (or “guess” or target value)** about the value of a parameter is tested by examining the evidence in the data. A **confidence interval** (also called an **interval estimate**) is a **range of values (or “guesses”)** within which we believe the true value of the parameter will fall based on the evidence in the data.

For any given sample of data, a confidence interval will provide more detailed information on the value of the parameter because it gives us an *entire range* of possible values for it. A hypothesis test results in a “yes/no” answer: it just tells us whether a particular value of the parameter that we chose to investigate is reasonable or not.

A hypothesis test is most useful when we just need support for a yes/no decision about a specific value of the parameter, and we don’t care much about all the other values that it *could* be. This often occurs with quality control situations where the process is supposed to be producing an item that meets certain specifications.

Example: Hypothesis Testing Situation

Suppose that we are working in quality control for a company that should be producing buckets of a certain chemical that weigh, on average, 32 pounds. If the machine is off—either filling the buckets with too little or too much of the chemical—the process needs to stop so that the filling machinery can be readjusted. We don’t really care *how* far off the process may be from 32 ounces on average; we may just care about whether it is or isn’t filling at a mean of $\mu = 32$ ounces. Thus, a hypothesis test (with an appropriate level of significance α that we would need to specify and summary information about the data) would be the most direct way to address this question.

Example: Confidence Interval Situation

Suppose a district manager for an insurance company wants to assess the performance of policies sold by representatives in her district by estimating the mean profit on a particular policy. The salespeople have some freedom to give discounts on policies for various reasons. She takes a random sample of 30 policies issued by these representatives and finds that the average profit on these policies was \$1439. The district manager doesn't have a particular, or "target," mean profit in mind; she just wants to determine a range of values within which the true mean profit will lie. She could then use this range of "guesses" (along with other information) to assess the performance of these sales representatives. The fact that a "range of guesses" for the parameter is the goal here makes the confidence interval approach the most direct way to address this question.

However, this can turn into a hypothesis testing situation fairly easily. Suppose that the company has stated that the mean profit on policies of this kind should be at least \$1500 in order for the company to meet its financial goals. If the mean profit is below this amount, the district manager will need to counsel the sales reps about what discounts they offer and how much the discounts are for. Now, there is a **target** to test: the company wants to see if the population mean profit $\mu < 1500$. Thus, the general manager would test the hypotheses $H_0: \mu \geq 1500$ versus $H_1: \mu < 1500$.

- Inference (intervals and tests) for means will **always** use the **Student's t** distribution.
- Inference (intervals and tests) for proportions will **always** use the **Z (standard normal)** distribution.
- Do you know the general relationship between confidence intervals and hypothesis tests?

In the case of testing a mean, here is the general rule:

If you calculate a $(1 - \alpha)100\%$ confidence interval for μ and a specific value of μ (say, μ_0) is NOT in the interval, then we would REJECT the two-sided null hypothesis at the α level.

For example, if I tell you that a 95% confidence interval for μ (that is, the population mean) is [12.1, 17.9], that interval represents the range of reasonable values, or "good guesses," for μ . If you were to test

$$H_0: \mu = 9$$

$$H_1: \mu \neq 9$$

you **would reject the null hypothesis** at the 0.05 level, since 9 is not in the interval.

However, if you were to test the following hypothesis:

$$H_0: \mu = 15$$

$$H_1: \mu \neq 15$$

you **would fail to reject the null hypothesis** at the 0.05 level, since 15 is in the interval.

The easiest way to remember this relationship is to think of a confidence interval as containing all the “good guesses” (for a given level of confidence) for the parameter. If you specify a value in a hypothesis test—i.e., you make a guess at a value—that is not among those guesses that are “good,” logic would tell you to reject the null hypothesis.

This same logic applies to all hypothesis tests. Intervals and tests are “two sides of the same coin.”

- Can you determine the form of the null and alternative hypotheses based on the context of the question?

Example:

A housing policy research group reported that 29% of homeowners in the U.S. struggle to pay monthly mortgage payments. A real estate company is interested in examining this claim in a particular U.S. state that has experienced some recent economic hardships. What would be a correct expression of the hypotheses if the company wants to see if the problem (in terms of the **proportion** of people struggling to pay) is worse in this particular state than in the U.S. as a whole? **Hint: Think about whether a large or small percentage would indicate “worse” conditions.**

Answer

Taking the view that a homeowner who struggles to pay monthly mortgage payments is a “bad” situation, if the proportion of people struggling to pay in a particular state is worse than the reported national percentage of 29%, then the alternative hypothesis would be

$$H_1: p > 0.29$$

and therefore, the null hypothesis would be

$$H_0: p \leq 0.29$$

Put together, the hypotheses for this test would be

$$H_0: p \leq 0.29$$

$$H_1: p > 0.29$$